# Ashisranjan Pradhan

me25mtech14016@iith.ac.in | ashisranjanpradhan53@gmail.com | +91 6370110364 | LinkedIn | GitHub

## EDUCATION

| Institution | Degree / Board | Passing Year | Score / CGPA |
|---|---|---|---|
| Indian Institute of Technology Hyderabad | B.Tech | 2025 | 8.00 |
| Adyant +2 Science College | Class XII (Council of higher secondary education, Odisha) | 2018 | 63.50% |
| Champeswar G P High school | Class X (Board of secondary education, Odisha) | 2016 | 84.83% |

## INTERNSHIPS

**Design Intern | Smart Knower (in association with Autodesk)**

*[01-12-2020 to 31-01-2021]*

- Learned and designed the combustion chamber of an Internal Combustion (IC) engine using SolidWorks and AutoCAD. Recognized with the "Best Intern Award" for exceptional execution and timeline adherence.

**Intern | KIIT School of Mechanical Engineering**

*[05-06-2021 to 04-07-2021]*

Learned and implemented Supply Chain Management principles, along with Lean and Six Sigma methodologies with Industrial case studies.

## SKILLS

**Languages and Frameworks:** Python, C, Java, MATLAB, C++, SQL

**Core CS**: Data Structures and Algorithms, Object-Oriented Programming, System Design Fundamentals

**Machine Learning:** Scikit-Learn, Reinforcement Learning, Deep Learning, Neural Network, Predictive modeling, Anomaly Detection, Clustering, Regression and Classification, Computer Vision

**Robotics & Control System:** ROS, PID and State-Feedback Control, State-Space modeling, LQR, Closed-loop Systems, Simulink

**Generative AI:** LLM Orchestration, LLM Application Integration, Prompt Engineering

**Data and Deployment:** Data Cleaning and wrangling, Data-driven Decision Making, Numpy, Pandas, Power BI, Plotly, SAP HANA Enterprise Integration, REST APIs, Git, Backend Automation, Firebase Authentication, Payment Gateway Integration

**Web Development:** Next.js, React, TypeScript, JavaScriptNode.js, Tailwind CSS, Prisma, Firebase

**Software Engineering Practices:** Software Development Life Cycle (SDLC), Agile, Unit Testing, Jenkins, Mavens, UNIX, Software Testing, Git / Version Control, Backend Automation, Code Integration and Review, Technical Documentation

## RELEVANT COURSES

Fundamentals of Software Engineering, Operating Systems, Relational Database Management System, Object Oriented System Design, Object Oriented Programming, Software Testing, Machine Learning and it's Applications, Control Systems Design, Fundamentals of Robotics, Computer Vision, Autonomous Robotics Systems, Marine Robotics, Vehicle Dynamics and Modeling, Probability and Optimization

## PROJECTS

**NIMBA Nature Cure | Team Lead -- Full-Stack & Backend (Patient Care Web Platform)**

- Led a development team building an end-to-end patient care web platform; supervised and integrated the overall client-server architecture and scaled the codebase for advanced runtime tasks.

- Directed design and integration of secure, token-based patient authentication using Firebase OTP, and supervised implementation of the payment gateway endpoints.

- Oversaw integration of Large Language Models (LLMs) into the application layer to power contextual, automated in-app features.

- Personally implemented the patient onboarding flow, patient portal UI components, and shared navigation/design-token system for responsive layouts.

*Tech: Next.js, React, TypeScript, Firebase, Tailwind CSS, Prisma* | *github.com/Ashisranjanpradhan/nimbanaturecure*

**Pro Trading | Institutional Quantitative & High-Frequency Trading Platform**

- Built an end-to-end, quantitative trading platform with a Level-2 Limit Order Book (LOB) microstructure engine, computing real-time micro-price, spread, and Order Flow Imbalance (OFI).

- Developed a spatial-temporal DeepLOB CNN-LSTM neural network (PyTorch) predicting short-term mid-price directional signals (Buy/Hold/Sell).

- Implemented a Deep Reinforcement Learning (PPO, Gymnasium) optimal execution simulator, benchmarking parent-child order liquidation strategies against TWAP/VWAP baselines.

- Built a Black-Scholes options risk matrix (Delta, Gamma, Theta, Vega) with 3D implied-volatility surface modeling, plus statistical arbitrage and Modern Portfolio Theory optimization engines.

- Designed a 6-stage onboarding flow with SHA-256 authentication, simulated KYC/SEBI risk profiling, live broker API integration (Angel One, Zerodha), and an AI Co-Pilot natural-language trade assistant; deployed live with a 14-test automated pytest suite.

*Tech: Python, Streamlit, PyTorch, Gymnasium, Node.js* | *Live: pro-trade.streamlit.app* | *github.com/Ashisranjanpradhan/lob-execution-engine*

**California Housing Price Prediction | Machine Learning**

- Built a regression pipeline predicting median house values across California districts using the California Housing dataset, progressing from a Decision Tree baseline to tuned ensemble models.

- Tuned a HistGradientBoosting Regressor via GridSearchCV, achieving the best performance (R-squared score of 0.840, Mean Absolute Error of 0.301), outperforming Random Forest and a single Decision Tree baseline.

*Tech: Python, Scikit-Learn, Jupyter Notebook* | *github.com/Ashisranjanpradhan/Real_estate_housing-price_prediction*

**Health Insurance Charges Prediction | Machine Learning**

- Built an end-to-end ML pipeline predicting individual medical insurance charges from demographic and health attributes, comparing Linear Regression, Random Forest, XGBoost, and a Neural Network (MLP).

- Tuned an XGBoost Regressor via GridSearchCV, reducing test RMSE by approximately 30.8% over the linear baseline; identified smoking status and BMI interaction as the dominant non-linear cost driver.

*Tech: Python, Scikit-Learn, XGBoost, Jupyter Notebook* | *github.com/Ashisranjanpradhan/Health_Insurance-ML*

**Australia Rain Prediction | Machine Learning**

- Built an end-to-end ML pipeline predicting next-day rainfall in Australia using the Kaggle Weather-in-Australia dataset, with a chronological train-validation-test split to prevent data leakage.

- Compared Logistic Regression and an Optimized Random Forest Classifier, achieving approximately 84-85% test accuracy, and analyzed recall trade-offs on the minority rain class under class imbalance.

*Tech: Python, Scikit-Learn, Jupyter Notebook* | *github.com/Ashisranjanpradhan/Australia_weather_prediction*

**Thesis Project: Bio-Mechatronic Smart Hoverboard | Systems & Controls Designer**

- Designing a bio-adaptive, self-balancing micro-mobility platform integrating wearable sEMG biosensors, a plantar

pressure (FSR) footplate array, and a 7-state Extended Kalman Filter for chassis state estimation.

- Developing a bio-adaptive 4-state LQR controller on an ARM Cortex-M7 (STM32H743ZI) platform that exploits the neuromuscular activation lead to proactively stabilize the platform ahead of physical chassis tilt.

- Authored the complete technical proposal covering system architecture, sensor-to-actuation integration flow, embedded firmware design, and treadmill-based experimental validation protocol.

## SCHOLASTIC ACHIEVEMENTS

### HONORS AND ACHIEVEMENTS

- Awarded "Best Performer of the Seminar by "STEELUN, NIT Rourkela" for analytical framework delivery.
- Honored with "Best Intern Award" by Smart Knower for exceptional execution of software endpoints and timeline adherence.
- 1st Place, State-Level Talent search Examination(2016).
- 1st Place, Block-Level Science Exhibition (2014).
- 1st place, Block-Level Seminar competition (2016).
- 2nd place, Block-Level Seminar competition (2015)

## PUBLICATIONS

N. D. Joshi, **Ashisranjan Pradhan**, C. Nakka, H. Allaka, A. K. Pandey, "Guidance and Navigation Control of Unmanned Surface Vehicle," *Proceedings of the ASME 2026 International Mechanical Engineering Congress and Exposition (IMECE2026)*, Chennai, India, Sept 3-5, 2026. Paper No. IMECE2026-188997.

## EXTRACURRICULAR

- Managed the annual company event at G.R Infraprojects Ltd, coordinating logistics, vendors, and cross-team participation
- Managed brand collaborations and sponsorships with multiple vendors and brands as part of the organizing team for KIIT FEST 2K19
- Cooking enthusiast and movie buff -- pursued as regular hobbies outside academics/work
- Volunteered in District-Level Mobile science exhibition in 2014.
- Participated in District Scout and Guide camporee 2015.

- Attended the MathWorks seminar "Bridging Industry Practice and Academic Innovation with Generative AI and Agentic AI": gained exposure to Generative AI and Agentic AI concepts and their application in industry and academic research.

## CERTIFICATIONS

### Advanced Python

Covered advanced language concepts including object-oriented design, data structures, file handling, and application-level scripting, forming the core programming foundation used across backend and data science projects.

### AutoCAD

Trained in 2D/3D computer-aided design and drafting, applied to mechanical component design including the combustion chamber design project at Smart Knower.

### C Programming

Covered core procedural programming, memory management, and algorithmic problem-solving, providing the foundation for Data Structures & Algorithms and systems-level software development.

## INTERESTS AND HOBBIES

- Cooking - regularly explores new recipes and cuisines.
- Movies - keeps up with a wide range of films and genres.
- Geopolitics and Global Finance -- follows macroeconomic trends and geopolitical developments and their impact on global financial markets.